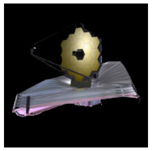

3248 - The Variable Stars in the ERS WLM Observations

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

We propose an archival analysis of the RR Lyrae population in the JWST ERS target WLM to develop and release tools for analysis of RR Lyrae with JWST. RR Lyrae observations have become commonplace components of nearby galaxy studies with the HST and the same should be true for JWST. However, JWST's observational mode provides challenges for RR Lyrae observations not encountered with the HST. These challenges include: (1) the inability to interleave observations with different filters, which can introduce aliasing and other sampling effects; (2) Due to data volume constraints, the JWST pipeline does not provide images suitable for variable star analysis and thus must be extracted through tailored reductions; (3) RR Lyrae variability amplitudes are smaller at longer wavelengths requiring new calibrations in JWST bandpasses. ERS observations of WLM provide an excellent sampling of RR Lyrae, but they are not being analyzed as part of the ERS program. We will compare the JWST RR Lyrae data with the archival HST observations of the same stars in order to produce a calibration of RR Lyrae with JWST filters that is anchored to Gaia observations. This program will provide the only guidance to date on how to extract and analyze extragalactic JWST observations of RR Lyrae. It will help to establish a foundation for the JWST Pop II extragalactic distance ladder.

OBSERVING DESCRIPTION

This is an archival proposal to analyze the ERS observations of the dwarf galaxy WLM.

The observations have already been obtained and are publicly available.